AE462 Lab 4: Transonic Airfoil Dynamics

Lab Session: Monday, March 2, 2026
Due: Monday, March 9, 2026, 9am CST

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This laboratory assignment investigates transonic airfoil aerodynamics using computational fluid dynamics (CFD). Simulations are conducted for the RAE 2822 airfoil under freestream conditions corresponding to Mach $M_\infty = 0.729$ and Reynolds number $Re = 6.5 \times 10^6$. The objectives of this study are to compare CFD predictions with experimental data, analyze airfoil performance across a range of angles of attack, and determine the flow conditions associated with a design lift coefficient of $C_l = 0.85$.

Introduction

Transonic airfoil aerodynamics involve complex flow phenomena including shock waves, compressibility effects, and shock-induced boundary layer separation. These phenomena strongly influence the aerodynamic performance of modern aircraft wings operating near the transonic regime.

In this lab, the RAE 2822 airfoil is simulated using the SU2 CFD solver with Reynolds-Averaged Navier–Stokes (RANS) turbulence modeling. The CFD results are compared against experimental measurements to assess the accuracy of the simulation. A lift polar is generated across multiple angles of attack, and the design condition corresponding to a target lift coefficient of $C_l = 0.85$ is analyzed.

I. Comparison to Experimental Data

A. Cp Distribution Comparison

Figure 1 provides a co-plot comparing the CFD-generated pressure coefficient distribution with the experimental data from Cook et al. for the RAE 2822 airfoil at $\alpha = 2.31°$.

B. Flow Modeling Assumptions

Discuss how the Cp distribution and flow physics would change under the following modeling assumptions.

1. Incompressible Viscous Flow ($M = 0.2$)

If the flow were instead modeled as incompressible viscous flow at $M = 0.2$, the compressibility effects associated with transonic flow would be absent. Most notably, the local supersonic pocket over the upper surface and the resulting shock wave would not appear in the solution. As a result, the Cp distribution would be much smoother and would not contain the sharp pressure jump associated with the shock. The leading-edge suction peak would still be present due to viscous flow over the airfoil, but it would generally be weaker than in the transonic case because the strong compressibility-driven acceleration of the flow would be missing. Overall, the pressure recovery would be more gradual, and the dominant transonic features of the flow would not be captured.

2. Inviscid Compressible Flow (Euler)

If the simulation were instead conducted using the inviscid compressible Euler equations, compressibility effects such as local supersonic acceleration and shock formation would still be captured. Therefore, the Cp distribution would still show a shock-related pressure jump on the upper surface. However, viscous effects would be missing, meaning that the boundary layer would not be resolved. This would eliminate important physical behavior such as viscous drag, shock-boundary-layer interaction, and any tendency toward shock-induced separation. In the Cp distribution, this would likely produce a sharper and more idealized shock structure with differences in the post-shock pressure recovery compared to the RANS solution. Thus, while the Euler model would preserve the major compressibility-driven features of the transonic flow, it would miss the viscous flow physics that strongly influence real airfoil performance.

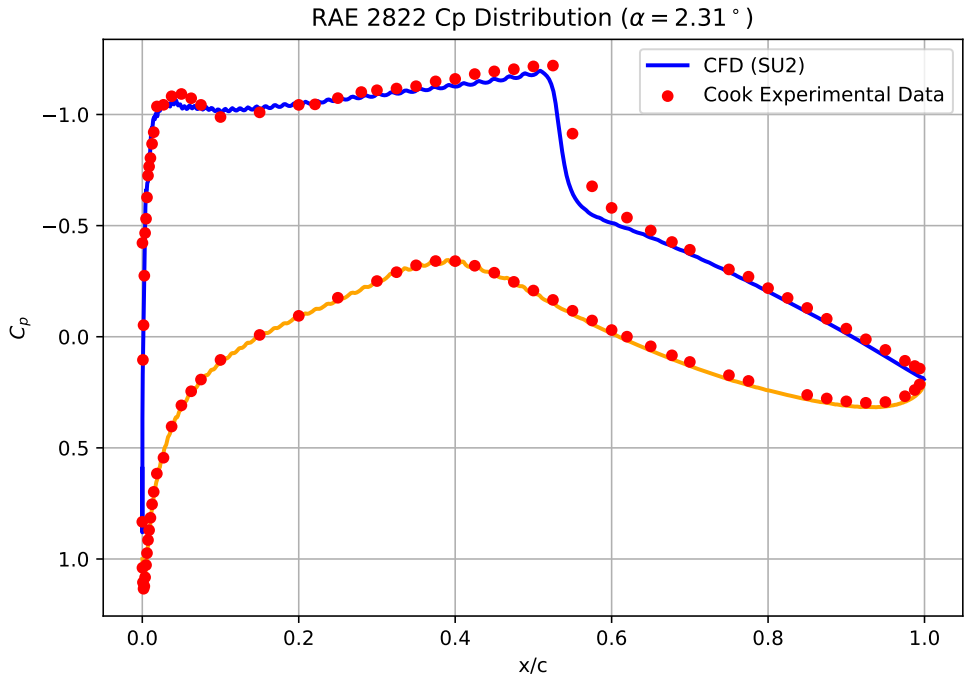


Fig. 1 Comparison of CFD and experimental pressure coefficient distributions for the RAE 2822 airfoil at $\alpha = 2.31°$.

II. Airfoil Performance Analysis

A. Lift Polar and Moment Coefficient

Figures 2 and 3 are the plots of the aerodynamic coefficients obtained from the CFD simulations.

B. Lift Curve Slope

Using the linear region of the lift polar between $\alpha = 0°$ and $\alpha = 4°$, compute the lift-curve slope:

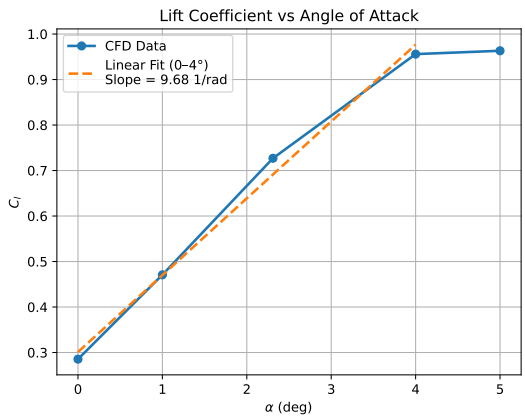
$$C_{l,\alpha} = \frac{dC_l}{d\alpha}$$

Compare this result to the theoretical compressible lift slope

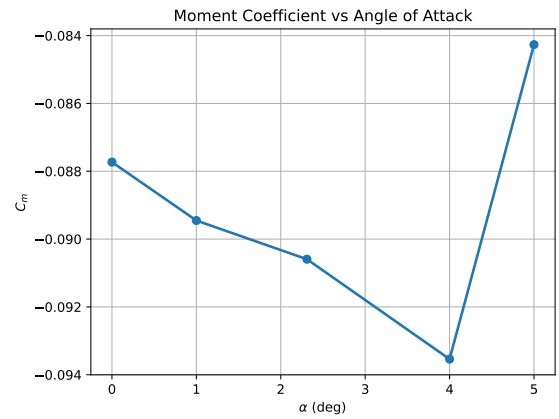
$$C_{l,\alpha} = \frac{2\pi}{\beta}, \qquad \beta = \sqrt{1 - M_\infty^2}$$

Calculated value: 9.68 rad^{-1}
Theoretical value: 9.18 rad^{-1}
Percent difference: 5.43%

The CFD result is slightly larger than the theoretical compressible lift slope. This difference is expected because the theoretical expression assumes an inviscid, infinitely thin airfoil with attached flow and does not account for viscous effects, airfoil thickness, or shock-related transonic flow behavior. In contrast, the CFD solution includes the actual RAE 2822 geometry, viscous effects through RANS turbulence modeling, and compressibility effects associated with transonic flow. These factors lead to small deviations from the idealized thin-airfoil prediction.



C_l **vs** α



C_m **vs** α

Fig. 2 Lift and moment coefficients as a function of angle of attack.

III. Airfoil Design Condition ($C_l = 0.85$)

A. Design Condition Summary

The design condition corresponds to a target lift coefficient of $C_l = 0.85$. Using the fixed-C_l mode in SU2, the solver adjusts the angle of attack until the specified lift coefficient is achieved. The resulting aerodynamic coefficients and angle of attack are summarized in Table 1.

Table 1 Airfoil aerodynamic coefficients at the design condition.

α (deg)	C_l	C_d	C_m
2.989	0.850	0.01997	0.0931

B. Cp Distribution at Design Condition

Figure 4 is a plot of the C_p distribution produced by the RAE 2822 airfoil at the design condition.

C. Mach Number Contours

Figure 5 are the mach contours around the RAE 2822 airfoil at the design lift condition.

Appendix

The github repo including the gmsh, Su2, and Paraview files as well as the python code used for post-processing, data analysis, and figure generation are available at: `https://github.com/nikitaaa27/AE462-git/tree/main/Lab4`

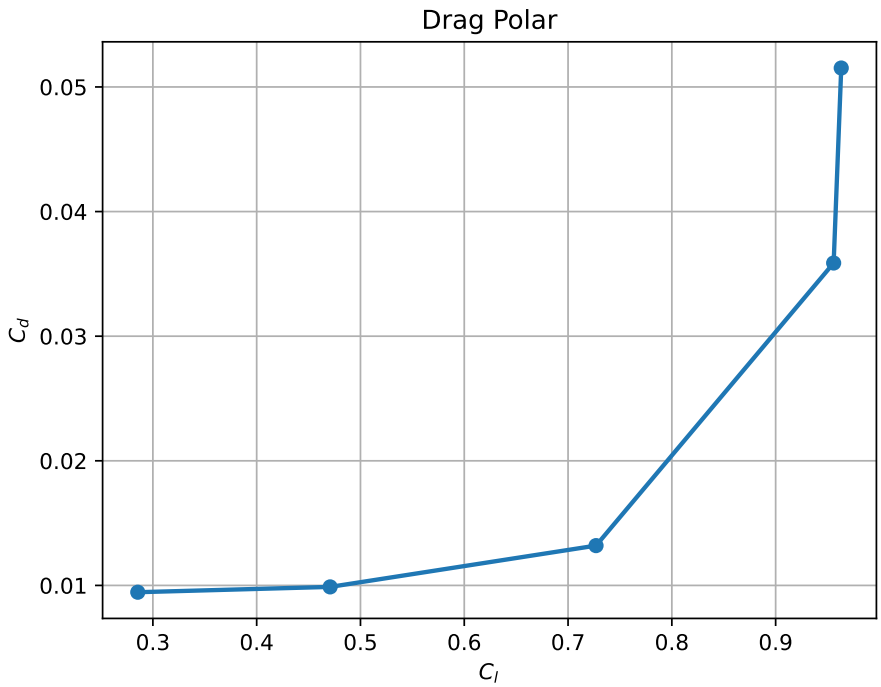


Fig. 3 Drag polar for the RAE 2822 airfoil.

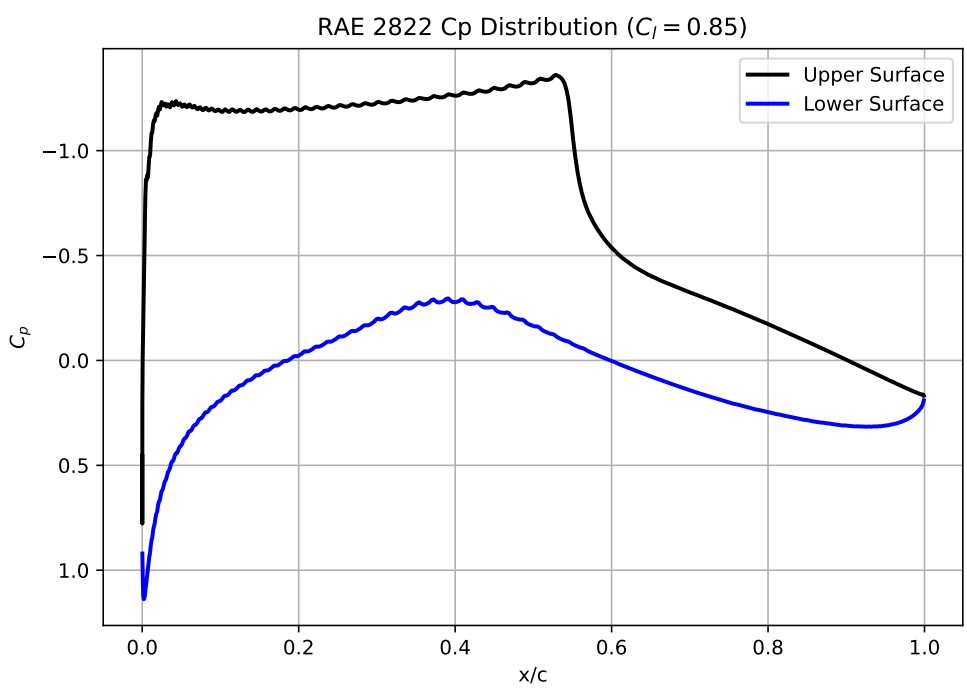
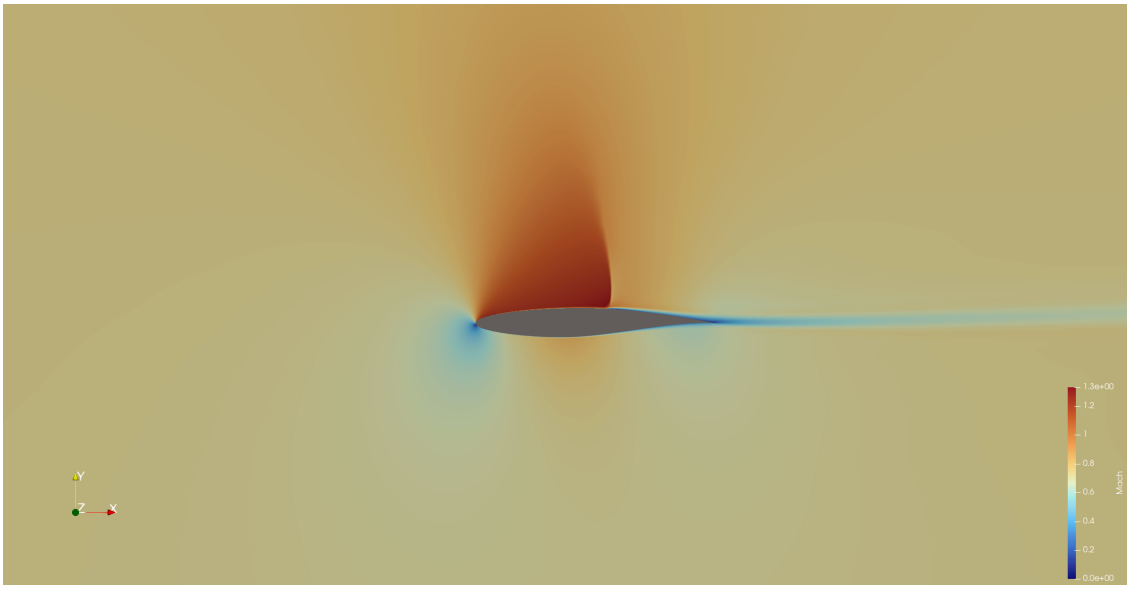
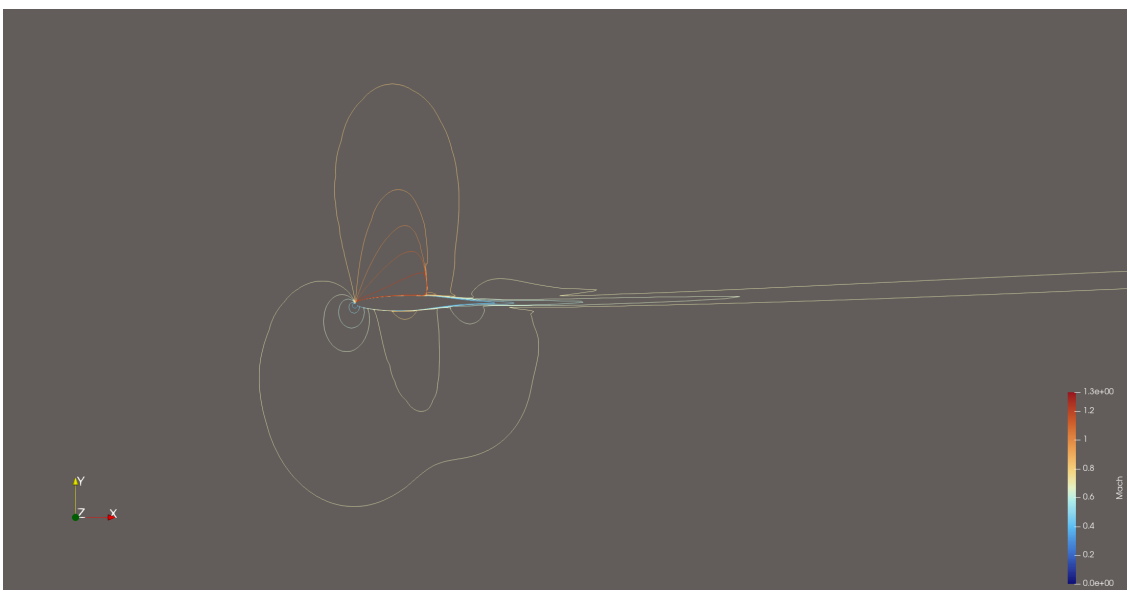


Fig. 4 Pressure coefficient distribution for the RAE 2822 airfoil at the design lift condition ($C_l = 0.85$).



(a) Mach number around the RAE 2822 airfoil at the design condition.



(b) Mach number isocontours highlighting the transonic shock structure.

Fig. 5 Mach number field around the RAE 2822 airfoil at the design lift condition ($C_l = 0.85$).